



Session 14A: Derivative Fundamentals — The Basic Toolbox

Phase 2 — Classical Techniques | 70 min

A derivative is instantaneous slope — how fast y changes when x twitches. You find it with a limit, but you compute it with rules. Memorize eight forms. Apply them in three steps: split, pull constants, match the dictionary.

Prerequisites: 13A (algebraic limits), 10A (exponents & logs), 11A (trig foundations)

Part A: What Is a Derivative — The Limit Definition

Example 1: The Limit Definition — Three Steps

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

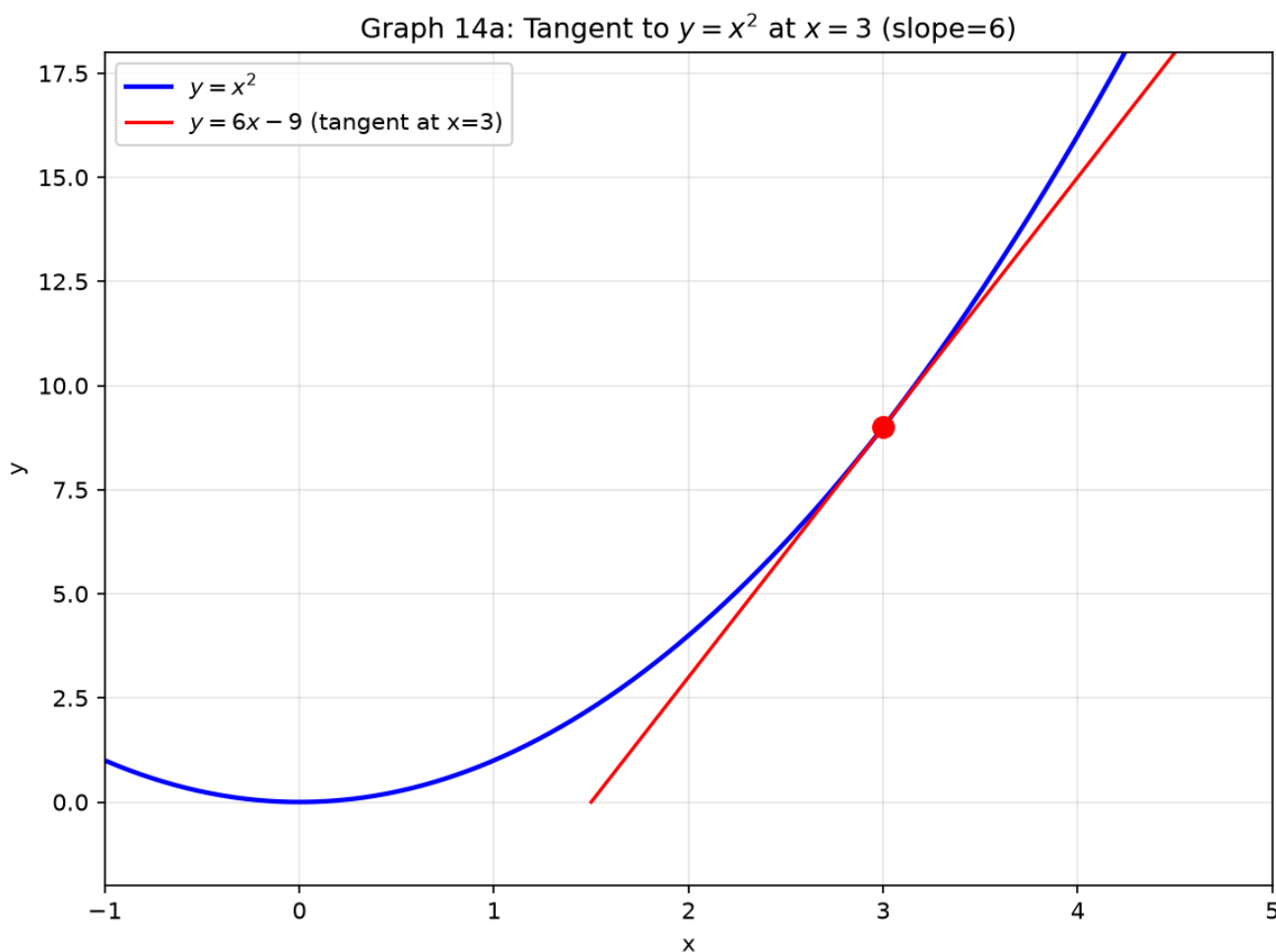
Procedure for computing $f'(a)$ from the definition:

Step	Action
1	Write the difference quotient: $\frac{f(a+h)-f(a)}{h}$
2	Simplify algebraically — expand, cancel, factor out h from numerator
3	Take the limit as $h \rightarrow 0$ — plug $h = 0$ into the simplified expression

$f(x) = x^2$ at $x = 3$:

- $\frac{(3+h)^2-3^2}{h} = \frac{9+6h+h^2-9}{h} = \frac{6h+h^2}{h}.$
- Factor out h : $\frac{h(6+h)}{h} = 6 + h$ (for $h \neq 0$).
- $\lim_{h \rightarrow 0} (6 + h) = 6$. So $f'(3) = 6$.

$f'(a)$ = slope of the tangent line at $x = a$.



Part B: The Derivative Dictionary — Eight Forms to Memorize

How to use this section: When you see any of these forms, apply the rule immediately. No thinking — just pattern-match.

Example 2: The Power Rule — $\frac{d}{dx} x^n = nx^{n-1}$

Procedure: Bring the exponent down as a coefficient. Subtract 1 from the exponent.

$f(x)$	$f'(x)$	Pattern
x^5	$5x^4$	Exponent 5 $\rightarrow$ coefficient, new exponent 4
x^{100}	$100x^{99}$	Same rule — no exceptions for large powers
$\sqrt{x} = x^{1/2}$	$\frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$	Fractional exponents work identically
$\frac{1}{x^3} = x^{-3}$	$-3x^{-4} = -\frac{3}{x^4}$	Negative exponents work identically
x	$1 \cdot x^0 = 1$	$x^1 \rightarrow 1 \cdot x^0 = 1$
7 (constant)	0	$7 = 7x^0$, derivative = $0 \cdot 7x^{-1} = 0$

The constant rule: $\frac{d}{dx} c = 0$. A flat line has zero slope everywhere.

The constant multiple rule: $\frac{d}{dx} (c \cdot f(x)) = c \cdot f'(x)$. The constant rides along.

Example 3: The Complete Derivative Dictionary

$f(x)$	$f'(x)$	Memory hook
x^n	nx^{n-1}	Exponent down, subtract 1
e^x	e^x	The only function that is its own derivative
a^x	$a^x \ln a$	Multiply by $\ln$ of the base
$\ln x$	$\frac{1}{x}$	Reciprocal of x (for $x > 0$)
$\log_a x$	$\frac{1}{x \ln a}$	$\frac{1}{x}$ divided by $\ln$ of base
$\sin x$	$\cos x$	$\sin \rightarrow \cos$ (no sign change)
$\cos x$	$-\sin x$	$\cos \rightarrow -\sin$ (sign FLIPS)

$f(x)$	$f'(x)$	Memory hook
$\tan x$	$\sec^2 x$	Also equals $1/\cos^2 x$

Three-step procedure for any derivative:

1. **Split** at every $+$ or $-$: $(f \pm g)' = f' \pm g'$.
2. **Pull** constants outside: $(cf)' = c \cdot f'$.
3. **Match** each piece to the dictionary above.

Example 4: Applying the Three-Step Procedure

$$\frac{d}{dx}(3x^4 - 2x^2 + 5x - 7).$$

Step	Action	Result
1	Split	$3x^4$ derivative + $(-2x^2)$ derivative + $(5x)$ derivative + (-7) derivative
2	Pull constants	$3 \cdot \frac{d}{dx} x^4 - 2 \cdot \frac{d}{dx} x^2 + 5 \cdot \frac{d}{dx} x - 7 \cdot \frac{d}{dx} 1$
3	Dictionary	$3(4x^3) - 2(2x) + 5(1) - 7(0) = 12x^3 - 4x + 5$

$$\frac{d}{dx}(2 \sin x + e^x - \ln x):$$

1. Split into three terms.
2. No constants to pull (coefficients stay).
3. Dictionary: $2 \cos x + e^x - \frac{1}{x}$.

Example 5: Exponentials and Logarithms — All Variants

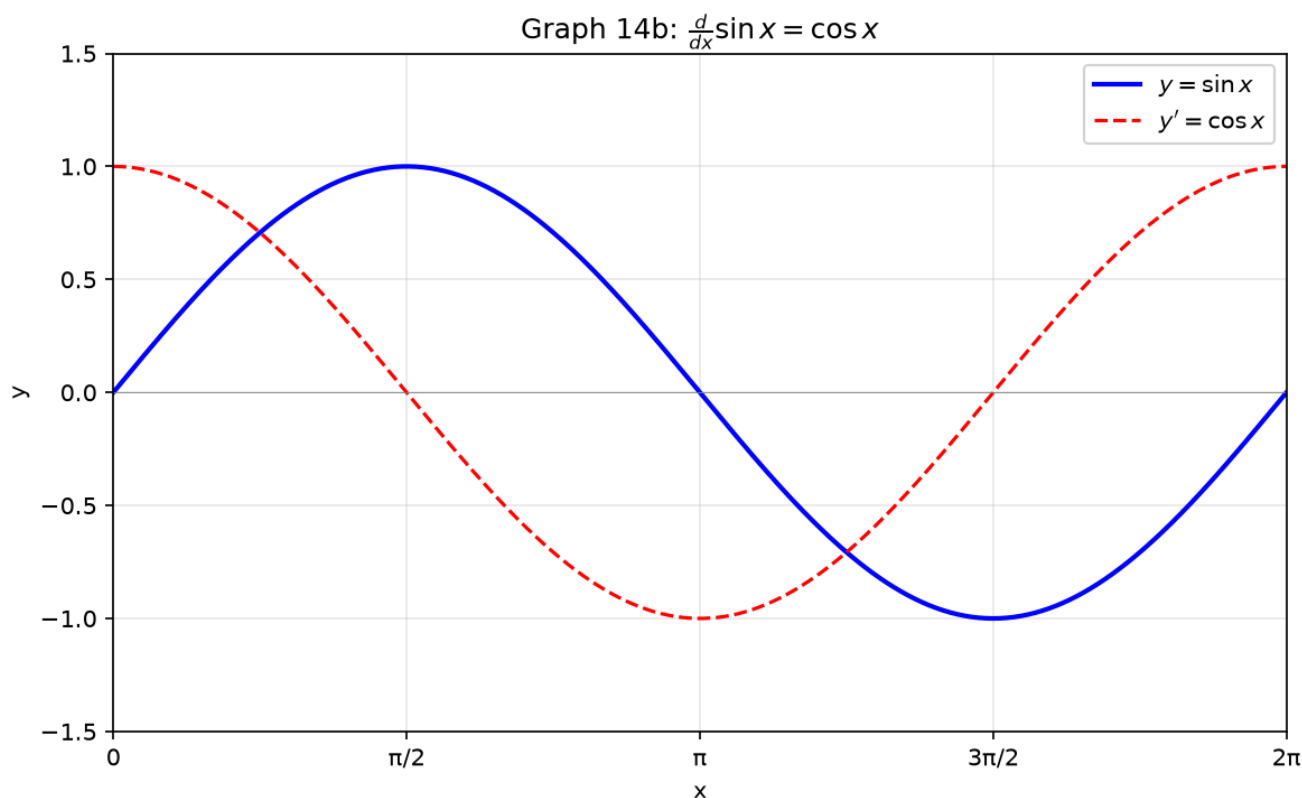
Form	Derivative	Note
e^x	e^x	Unchanged. The identity function of differentiation.
e^{kx}	ke^{kx}	Preview of chain rule (14B). For now, treat as a pattern.
a^x	$a^x \ln a$	$\ln a$ is just a constant multiplier.
$\ln x$	$1/x$	Domain: $x > 0$.
$\ln(kx)$	$1/x$	$\ln(kx) = \ln k + \ln x$, derivative of constant is 0.

$$\frac{d}{dx} 2^x = 2^x \ln 2. \quad \frac{d}{dx} \log_3 x = \frac{1}{x \ln 3}.$$

Example 6: Trigonometric Derivatives — The Cyclic Pattern

$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f^{(4)}(x)$
$\sin x$	$\cos x$	$-\sin x$	$-\cos x$	$\sin x$ (back!)
$\cos x$	$-\sin x$	$-\cos x$	$\sin x$	$\cos x$ (back!)

The cycle: $\sin \rightarrow \cos \rightarrow -\sin \rightarrow -\cos \rightarrow \sin$. Four derivatives return you to the start.



Part C: The Tangent Line — From Derivative to Geometry

Example 7: Writing the Tangent Line Equation — Three Steps

Procedure: Given $f(x)$ and a point $x = a$:

Step	Action
1	Compute $y_0 = f(a)$ — the point of tangency is $(a, f(a))$
2	Compute $m = f'(a)$ — the slope at that point

Step	Action
3	Write $y - y_0 = m(x - a)$ — point-slope form

$f(x) = x^2 + \ln x$ at $x = 1$:

1. $y_0 = f(1) = 1^2 + \ln 1 = 1 + 0 = 1$. Point: $(1, 1)$.
2. $f'(x) = 2x + \frac{1}{x}$. $m = f'(1) = 2 + 1 = 3$.
3. $y - 1 = 3(x - 1) \rightarrow y = 3x - 2$.

Example 8: Finding Horizontal Tangents

Procedure: Set $f'(x) = 0$ and solve for x .

$f(x) = x^3 - 3x^2 - 9x + 5$. $f'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x - 3)(x + 1)$.

$f'(x) = 0$ at $x = 3$ and $x = -1$. These are the x -coordinates where the tangent is horizontal.

Up to here: Derivative = limit of difference quotient. Dictionary of 8 forms. Three-step procedure: split $\rightarrow$ pull constants $\rightarrow$ match dictionary. Tangent line: find point, find slope, write equation.

Common Mistakes

Mistake 1: Using the power rule on exponentials

Wrong: $\frac{d}{dx} e^x = x e^{x-1}$. **Right:** $\frac{d}{dx} e^x = e^x$. The power rule is ONLY for x^n (variable base, constant exponent). e^x has constant base, variable exponent — completely different.

Mistake 2: Forgetting the negative sign for $\cos x$

Wrong: $\frac{d}{dx} \cos x = \sin x$. **Right:** $\frac{d}{dx} \cos x = -\sin x$. Differentiate twice and you'll see why — $\sin \rightarrow \cos \rightarrow -\sin \rightarrow -\cos \rightarrow \sin$.

Mistake 3: $\frac{d}{dx} \ln x = x \ln x$ or $\ln x$

Wrong: Confusing derivative with antiderivative. **Right:** $\frac{d}{dx} \ln x = \frac{1}{x}$. The derivative of $\ln x$ is simple; the integral of $\ln x$ is $x \ln x - x + C$ (Session 16B).

Mistake 4: Forgetting to set $f'(x) = 0$ for horizontal tangents

Wrong: "The minimum of x^2 is at $x = 0$ because I can see it on the graph." **Right:** Set $f'(x) = 2x = 0 \rightarrow x = 0$. The derivative locates extrema precisely, without guessing from a graph.

What We Just Did

- (1) Limit definition: $f'(a) = \lim_{h \rightarrow 0} [f(a+h) - f(a)]/h$.
Expand, cancel h , plug $h=0$.
- (2) Derivative dictionary (8 forms): $x^n \rightarrow nx^{n-1}$, $e^x \rightarrow e^x$, $\ln x \rightarrow 1/x$,
 $\sin x \rightarrow \cos x$, $\cos x \rightarrow -\sin x$, $\tan x \rightarrow \sec^2 x$.
- (3) Three-step procedure for any derivative:
Split at $\pm$ $\rightarrow$ Pull constants out $\rightarrow$ Match each piece to the dictionary.
- (4) Tangent line: point $(a, f(a))$, slope $f'(a)$, equation $y - f(a) = f'(a)(x - a)$.

Practice 1

Use the limit definition to find $f'(2)$ for $f(x) = x^2 + 3x$.

$\rightarrow$ Reference: **Example 1**

Solutions: [Solutions](#)

Practice 2

Differentiate $f(x) = 4x^5 - 3x^3 + 2x - 1 + \frac{1}{x}$. Use the 3-step procedure: split, pull constants, match dictionary.

→ Reference: **Example 2, 4**

Solutions: [Solutions](#)

Practice 3

Differentiate $g(x) = 3e^x - 2 \ln x + 5 \sin x - \cos x$.

→ Reference: **Example 3, 4, 6**

Solutions: [Solutions](#)

Practice 4

Differentiate $h(x) = 2^x + \log_3 x + \tan x$.

→ Reference: **Example 5, 6**

Solutions: [Solutions](#)

Practice 5

Find all x where the tangent line to $f(x) = x^3 - 3x^2 - 9x + 5$ is horizontal.

→ Reference: **Example 8**

Solutions: [Solutions](#)

Practice 6: Real Battle

Find the tangent line to $f(x) = x^2 + \ln x$ at $x = 1$. Write your answer in $y = mx + b$ form.

→ Reference: **Example 7**

Solutions: [Solutions](#)

Basic Algebra Drill — Derivative Fundamentals (10 Problems)

Run the 3-step procedure: split, pull constants, match dictionary.

- D1.** $\frac{d}{dx}(7x^4)$. Power rule: exponent down, subtract 1.
- D2.** $\frac{d}{dx}(-3x^{10})$. Negative coefficient rides along.
- D3.** $\frac{d}{dx}(\sqrt[3]{x})$. Write as $x^{1/3}$ first.
- D4.** $\frac{d}{dx}(5e^x)$. Constant multiple: 5 stays, $e^x \rightarrow e^x$.
- D5.** $\frac{d}{dx}(4 \ln x)$. Constant multiple: 4 stays, $\ln x \rightarrow 1/x$.
- D6.** $\frac{d}{dx}(3 \sin x - 2 \cos x)$. Split, constant multiple, dictionary.
- D7.** $\frac{d}{dx}(\tan x + \sec x)$. Dictionary: $\tan x \rightarrow \sec^2 x$, $\sec x \rightarrow \sec x \tan x$.
- D8.** $\frac{d}{dx}(2^x + \log_5 x)$. General forms: $a^x \rightarrow a^x \ln a$, $\log_a x \rightarrow 1/(x \ln a)$.
- D9.** $\frac{d}{dx}\left(\frac{1}{x^4}\right)$. Rewrite as x^{-4} , apply power rule.
- D10.** Find $f'(0)$ for $f(x) = x^3 - 2x^2 + 5x - 1$. Differentiate, then plug $x = 0$.

Solutions: [Solutions](#)

Advanced Algebra Drill — Derivative Fundamentals (10 Problems)

Prove, connect, and extend the basic rules.

- A1.** Use the limit definition to prove $\frac{d}{dx}x^2 = 2x$. Show all three steps.
- A2.** Find a and b so $f(x) = ax^2 + bx$ has $f'(1) = 5$ and $f'(2) = 9$. Set up equations, solve.
- A3.** Differentiate $f(x) = \frac{x^3}{\sqrt{x}}$. Simplify to $x^{5/2}$ first, then apply power rule.
- A4.** Find the point on $y = x^2$ where the tangent line has slope 6. Set $2x = 6$.
- A5.** Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x}$. (Common denominator trick.)
- A6.** Find all x where $f(x) = x^3 - 6x^2 + 9x$ has horizontal tangent lines. Factor $f'(x)$.
- A7.** Prove $\frac{d}{dx} \tan x = \sec^2 x$ using $\tan x = \frac{\sin x}{\cos x}$ and the quotient rule.
- A8.** Differentiate $f(x) = \ln(x^2)$ two ways: (a) $\ln(x^2) = 2 \ln x \rightarrow 2/x$. (b) Chain rule. Compare.
- A9.** A function satisfies $f'(x) = f(x)$ for all x and $f(0) = 3$. What is $f(x)$? (Which function equals its own derivative?)
- A10.** Find the tangent line to $y = e^x$ that passes through the origin $(0, 0)$. The line has slope e^a , passes through (a, e^a) and $(0, 0)$. Solve for a .

Solutions: [Solutions](#)

Today's Procedure

When you see...	Do this...
x^n (any $n \neq 0$)	Power rule: nx^{n-1}
e^x	Unchanged: e^x

When you see...	Do this...
a^x	Multiply by $\ln a$: $a^x \ln a$
$\ln x$	Reciprocal: $1/x$
$\sin x, \cos x, \tan x$	Dictionary: $\cos x, -\sin x, \sec^2 x$
Sum/difference of functions	Split at $+/ -$, differentiate each
Constant times a function	Pull the constant out
Need slope at $x = a$	Compute $f'(a)$
Need tangent line at $x = a$	Point $(a, f(a))$, slope $f'(a)$, equation $y - f(a) = f'(a)(x - a)$
Need horizontal tangents	Set $f'(x) = 0$, solve for x

Terminology

What we call it	Math term	Notation
instantaneous slope	derivative	$f'(x), \frac{dy}{dx}$
limit of difference quotient	derivative definition	$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
exponent down, subtract 1	power rule	$\frac{d}{dx} x^n = nx^{n-1}$
its own derivative	exponential invariance	$\frac{d}{dx} e^x = e^x$
touch at one point	tangent line	$y - f(a) = f'(a)(x - a)$
slope equals zero	horizontal tangent	$f'(x) = 0$